

Muhammad Raz

Forward Deployed Engineer & Full-Stack Developer | Cloud Architecture | AI Integration

Kannur, Kerala, India

9605633156

| raazraz621@gmail.com

| [LinkedIn](#)

| [GitHub](#)

| [Portfolio](#)

Technical & Professional Expertise

Languages: TypeScript, JavaScript, Python, SQL, Dart

Backend & Cloud: Node.js, Express, FastAPI, Supabase, Deno Edge Functions, AWS Rekognition

AI & LLM Orchestration: LangChain, LangGraph, LangSmith, Vector DBs, RAG Pipelines

Frontend & Mobile: React.js, Vite, Next.js, Tailwind CSS, Flutter, PWA integration

Data & Architecture: PostgreSQL (RLS, pg_cron), MongoDB, Redis, Distributed Caching, Event-Driven OOP

DevOps & Consulting: Docker, CI/CD, Git, Client Diagnostics, Operations Research, Rapid Prototyping

Professional Experience

Forward Deployed Engineer (FDE)

Aug 2025 – Present

Emilda Solutions

Calicut, India

- **Rapid Technical Deployment:** Embedded directly with enterprise clients to architect and ship scalable web, mobile, and edge-computing solutions, successfully delivering end-to-end platforms within strict deadlines.
- **Client Diagnostics & Solutions:** Acted as the primary technical bridge for business stakeholders, diagnosing complex on-premise operational bottlenecks and replacing them with automated, high-ROI software ecosystems.
- **Applied AI/LLM Integration:** Forward-deployed generative AI capabilities into live client environments, building custom orchestration pipelines and autonomous diagnostic agents using LangChain, LangGraph, and LangSmith.

Software Engineer

Sep 2024 – Jul 2025

White Matrix

India

- **Cross-Platform Development:** Engineered performant Flutter mobile applications, seamlessly integrating complex UI features with resilient Express.js backend microservices within an agile team.
- **Quality Assurance & Delivery:** Enforced rigorous CI/CD and version control protocols, resolving critical architectural bottlenecks to ensure rapid and stable production deployments.

Key Engineering Projects

Rozana Procurement Intelligence Engine

2026

Python, LangGraph, LangChain, Supabase (PostgreSQL), FastAPI, Claude Haiku

- **AI Architecture Design:** Architected a deterministic math engine and a 7-node LangGraph decision graph to replace gut-feel purchasing with a data-driven advisory system.
- **Bounded LLM Narration:** Confined AI integration (Claude Haiku via OpenRouter) strictly to a narration layer that explains pre-computed math decisions in plain English, ensuring 100% reproducible and auditable outputs.
- **Data Ingestion & Backtesting:** Processed 893,437 sales rows from a Pearl D3 ERP into Supabase and backtested against 2,350 historical purchases, successfully identifying SAR 642,382 in preventable over-buys.
- **Infrastructure Resilience:** Resolved a severe Supabase disk-full crash loop caused by 1-2 million unbounded parallel API calls by enforcing 500-1000 row batch writes and strict concurrency caps.
- **Approval Workflow System:** Built a 15+ endpoint FastAPI backend to serve the recommendations through a human-in-the-loop approval workflow, fully traced end-to-end using LangSmith.

Station Command — Fuel Station Operations Platform

2026

React, TypeScript, Supabase (PostgreSQL, Edge Functions), Python, AWS Rekognition

- **End-to-End Delivery:** Shipped a role-based PWA operations platform for an HPCL fuel station, replacing paper workflows for a 15-person team, owning full-stack architecture from hardware integration to deployment.
- **Hardware Data Ingestion:** Engineered a Python sync agent on embedded Linux POS terminals, streaming 1,700+ daily transactions to PostgreSQL via Edge Functions, backfilling 6 years of data using idempotent upserts.
- **Resilience Engineering:** Resolved critical sync stalls by re-architecting the ingestion cursor to a timestamp-based bounded drain loop with self-healing restarts, ensuring 100% data pipeline uptime.
- **Enterprise Security:** Implemented Supabase email-OTP auth, RBAC, and Row-Level Security (RLS) across 30+ tables with SECURITY DEFINER RPCs, executing the migration with zero downtime.
- **Commercial Scalability:** Architected and packaged the backend infrastructure as a configuration-driven, multi-tenant module, successfully driving the platform's rollout to additional enterprise client sites.

Education

Bachelor's Degree in Computer Application (BCA)

Jamia Hamdard University

Delhi, India